



Elevational differentiation in metabolic cold stress responses of an endemic mountain tree



Karin Schrieber^{a,b,*,1}, Yolanda Cáceres^{c,d,1}, Alicia Engelmann^b, Paula Marcora^e, Daniel Renison^f, Isabell Hensen^{c,d}, Caroline Müller^b

^a Kiel University, Institute for Ecosystem Research, Geobotany, 24118, Kiel, Germany

^b Bielefeld University, Faculty of Biology, Department of Chemical Ecology, Bielefeld, Germany

^c Martin-Luther-University Halle-Wittenberg, Institute of Biology, Geobotany & Botanical Garden, Halle (Saale), Germany

^d German Centre for Integrative Biodiversity Research (iDiv) Halle-Jena-Leipzig, Leipzig, Germany

^e National University of Cordoba, Institute for Multidisciplinary Plant Biology (IMBIV), Córdoba, Argentina

^f National Scientific and Technical Research Council, Instituto de Investigaciones Biológicas y Tecnológicas, (IIBYT - CONICET – Universidad Nacional de Córdoba), Córdoba, Argentina

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ABSTRACT

The decline of mountain forests under global change is alarming and conservation is challenged by the need to assemble populations able to deal with the predicted environmental shifts. Assessing intraspecific differentiation in stress susceptibility of tree species along elevational gradients is crucial to identify adequate genetic sources for mountain forest restoration. Here, we assessed injury and metabolic responses across cold stress intensity gradients in low and high elevation populations of the endangered mountain tree *Polylepis australis*. We hypothesized that populations from high elevational origins will show weaker increases of injury and more pronounced metabolic responses with rising cold stress intensity than low elevation populations. The expected differentiation among elevational origins was not observed for cold stress injury, but for metabolic stress responses. Plants responded to cold with enhanced concentrations of sugars, organic acids, amines and specialized metabolites with freezing point depressing, cryo-protective and anti-oxidative properties. As predicted, these responses were more pronounced and partially increased more strongly with stress intensity in high than low elevation populations, indicating local adaptation to a higher frequency and intensity of cold events at high elevations. We discuss the power of metabolomics approaches in unravelling intraspecific differentiation in tree stress resistance across small geographic scales as well as implications for future research on the evolution of cross-resistance to multiple stressors and for forest restoration.

1. Introduction

Low temperature is a key abiotic environmental stressor limiting the latitudinal and elevational distribution of tree species (Charrier et al., 2015). The susceptibility of trees to cold stress varies among species and populations within species (Strimbeck et al., 2015). Intraspecific variation has been detected across broad latitudinal ranges, where geographic isolation prevents genetic exchange among tree populations and facilitates trait differentiation via genetic drift or selective processes (Savolainen et al., 2007; Kreyling et al., 2014). Population divergence in cold susceptibility may also occur across smaller geographic scales, such as mountain ranges, which exhibit steep climatic gradients with decreasing temperatures as physiologically most relevant changes

occurring towards the summits (Körner, 2007). The few studies addressing this issue indeed found support that the increased frequency and intensity of cold stress events towards high elevations selects for phenological traits, such as late flushing and short growing season, to enhance individual fitness in deciduous trees (Inouye, 2008; Vitasse et al., 2009, 2013).

Next to phenological adaptations, adjustments in plant metabolism can contribute substantially to cold stress resistance (Guy et al., 2007; Heidarvand and Maali, 2010; Janská et al., 2010), which can be expected particularly for evergreen mountain tree species that expose fully differentiated leaves to frost. Nevertheless, metabolic cold stress responses and population differentiation in these complex traits across elevational gradients remains largely unexplored for such tree species

* Corresponding author at: Kiel University, Institute for Ecosystem Research, Department of Geobotany, Olshausenstr. 75, 24118, Kiel, Germany.

E-mail address: kschrieber@ecology.uni-kiel.de (K. Schrieber).

¹ Contributed equally to this work.

(but see Rada et al., 2001). Research on predominantly herbaceous model plants (reviewed in Sakai and Larcher, 1987; Guy et al., 2007; Heidarvand and Maali, 2010; Janská et al., 2010) established that freezing damages living cells via the formation of extracellular ice crystals, which is associated with cell dehydration and shrinkage. The concentration of ions under water deficit destabilizes bio-membranes and causes the denaturation of macromolecules. Moreover, photosystems can incur severe damage from increased levels of reactive oxygen species (ROS) following the reduced consumption of light-absorbed energy during CO₂ fixation under frost drought. Metabolic responses to these physiological dysfunctions comprise shifts in fatty acid composition to maintain the liquid crystalline state of lipid membranes as well as the synthesis of antioxidants. Moreover, plants can accumulate multiple cryo-protectors (e.g., sugars, sugar alcohols, organic acids, amines, amino acids) that mediate freezing point depression, stabilize lipid membranes, maintain the conformation of proteins and dilute toxic ion concentrations. In tree species, these changes can be comprehensively addressed with untargeted metabolomics approaches that require no prior knowledge on the physiology of the study species (Jones et al., 2013; Lankadurai et al., 2013).

Information on population differentiation in tree cold stress susceptibility across elevational gradients can have tangible implications for mountain forest conservation. Many mountain forests currently experience ongoing degradation, mainly as a result of land use and climate change (Sasaki and Putz, 2009; Fort, 2015). Understanding the patterns of variation in cold stress susceptibility among populations can be critical for accurately predicting the impacts of human-mediated environmental change on natural recruitment (Lindner et al., 2010). Given their ecological, economic and socio-cultural value, degraded mountain forests are partially restored in elaborate conservation programs (Rocca et al., 2014; Seidl et al., 2019). Information on

intraspecific variation in cold stress resistance is required to identify adequate genetic sources for these measures (Broadhurst et al., 2008; Cochrane et al., 2015). If tree species exhibit elevational population differentiation, their seeds should be established at elevations corresponding to their origin to prevent severe injury following cold events and ensure the sustainability of restoration measures (Whiteley et al., 2015).

In this study, we explore population differentiation in cold stress resistance and the associated metabolic signature across the elevational distribution range of *Polylepis australis* Bitter (Rosaceae), an endemic evergreen tree locally dominating the mountain forests of Argentina. During the past century, fires and intensive livestock rearing have forced this keystone tree species back to steep ravines (Renison et al., 2006; Sparacino et al., 2019). Anthropogenic activities still tend to prevent natural forest re-colonization out of these protected refugia and thus maintain grasslands in areas with potential for forest development causing extensive soil erosion processes (Cingolani et al., 2013; Marcora et al., 2013). Current conservation projects require information on variation in stress resistance to improve seed sourcing for the broad scale restoration of these *P. australis* forests (Renison et al., 2005). We thus exposed *P. australis* saplings of populations from low and high elevational origin to four experimental treatments with increasing cold stress intensities in two experiments. In experiment 1, the stress intensity gradient involved lethal negative temperatures for the assessment of cold injury. In the second experiment, the gradient comprised only sub-lethal negative temperatures for the assessment of leaf metabolic cold stress responses. We hypothesized that populations from high elevational origins will show weaker increases of injury and more pronounced metabolic responses with rising cold stress intensity than low elevation populations.

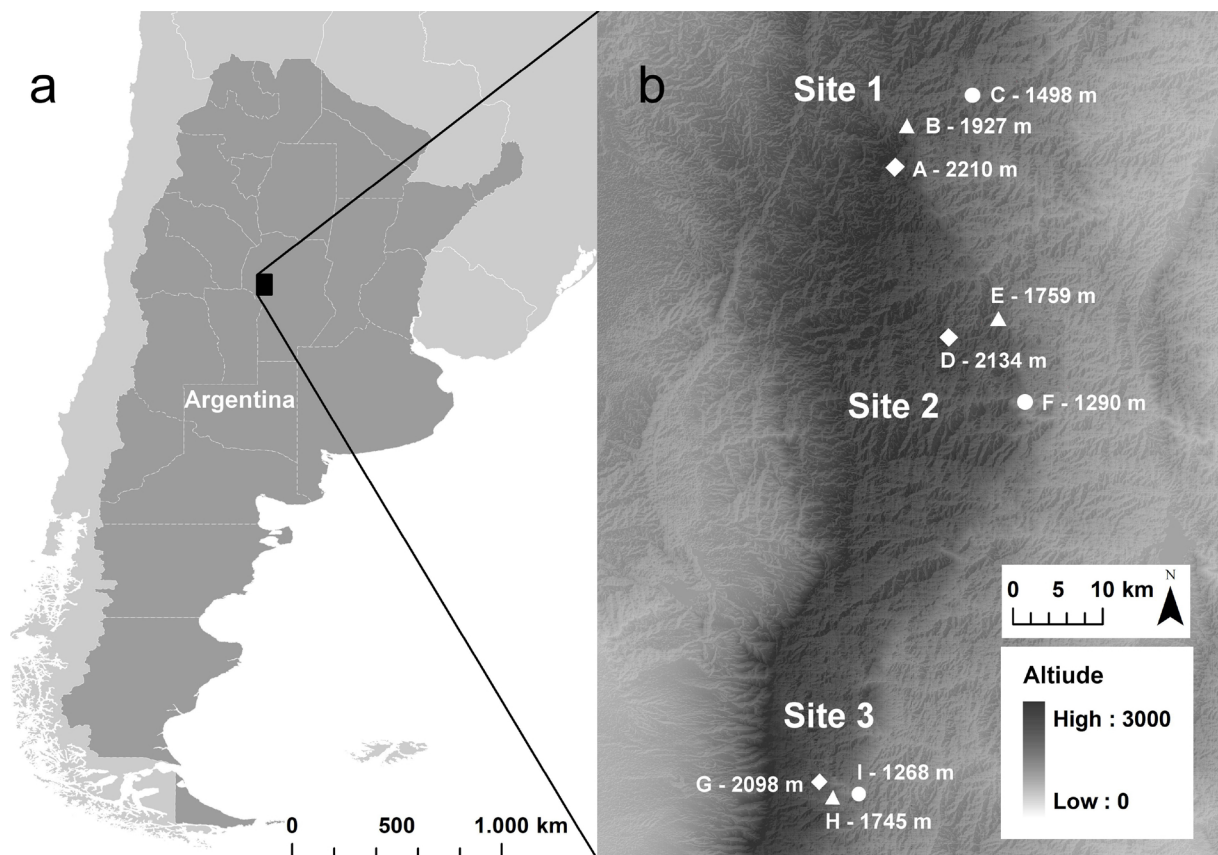


Fig. 1. Study area. Map of the geographic location of *P. australis* populations (A–I) sampled at low (circles), middle (triangles) and high (rhombuses) elevational levels at each of three sites (1–3).

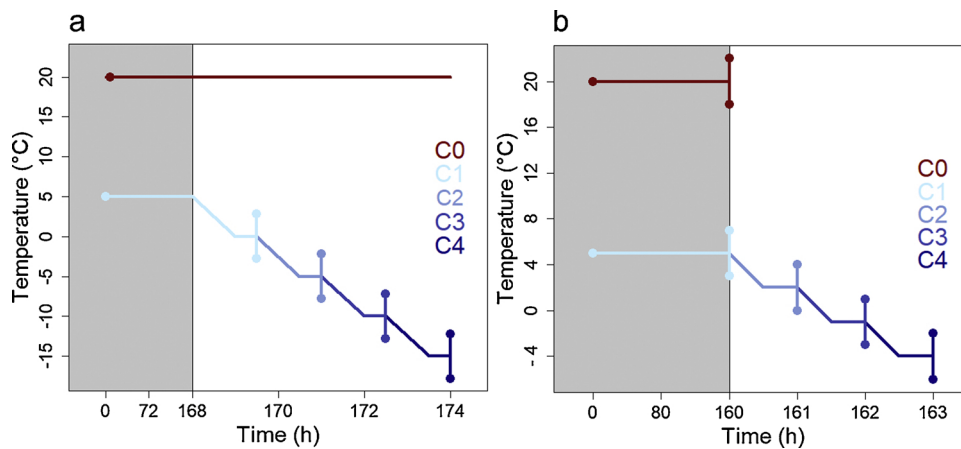


Fig. 2. Experimental cold stress treatments. Temperature-time course of experimental treatments (CO–C4) for the assessment of **a)** cold stress injury (experiment 1) and **b)** metabolic cold stress responses (experiment 2) in *P. australis*. The transition between grey and white background marks a change in the time axis scaling as well as the transfer of C1–C4 plants **(a)** or C2–C4 plants **(b)** from the climate cabinet to the cold chamber. Vertical bars represent the time point when plants designated to a specific cold stress intensity treatment were removed from the cold chamber and transferred to benign conditions to assess damage and mortality **(a)** or when plants were harvested for chemical analyses **(b)**.

2. Material & methods

2.1. Study species and area

Polylepis australis Bitt. (Rosaceae) is an evergreen tree endemic to Argentina, where it dominates the canopy of the local high elevation forest-shrub lands. This keystone species provides important mutualistic interactions with numerous other endemic species (Gareca et al., 2010) and is used to reduce soil erosion through forest restoration in degraded landscapes (Renison et al., 2017). *Polylepis australis* occurs in two disjunct regions: the eastern Andean slope of North-Western Argentina and the Córdoba mountains in Central Argentina. Our study area was located in the latter region (Fig. 1a), which is subjected to a subtropical/meso-thermal climate with dry winters (Peel et al., 2007). The elevational distribution of *P. australis* in this region covers 900 (low extreme) to 2600 (high extreme) m asl, whereby the frequency and intensity of frost events increases from the mountain foot towards the summit (Pollice, 2016; Supporting information Table S1).

2.2. Field sampling and plant propagation for the experiments

In early 2017, we collected seed material from isolated and sufficiently large (min. 70 mature individuals) *P. australis* forest patches. We walked the elevational gradient in each of three sites separated by at least 20 km and harvested seeds at low (1200–1500 m), middle (1700–1900 m) and high (2100–2200 m) elevational levels, which were separated by at least 3 km, resulting in a total number of nine *P. australis* populations (Fig. 1, Supporting information Table S2). We intentionally excluded very small populations occurring at elevation extremes because such range edge stands are typically characterized by low genetic diversity (Sexton et al., 2009) and not in the focus of restoration measures. Approximately 2000 seeds were hand-harvested from each of 15 to 20 vigorous individuals (uninjured, dense foliage with healthy green leaves) per population, which were haphazardly selected. Collected seeds were cleaned, pooled within populations and stored in paper bags at room temperature.

In October 2017, seeds from all populations were sown individually into pots (7 cm diameter) containing a compost : sand (2 : 1) mixture and germinated in a greenhouse at 20 °C/10 °C in a 12 h light/12 h dark cycle. Vital seedlings were further cultivated to the desired developmental sapling stage, watered when necessary and weekly randomized.

2.3. Experiment 1: cold stress injury

At the age of 90 days (February 2018), we assessed sapling leaf damage and mortality under a gradient of increasing cold stress intensity in *P. australis* populations from low, mid and high elevations. Based on the availability of saplings (Supporting information Table S2),

we exposed seven to 15 individuals per site × elevational origin combination to five experimental treatments (control [CO], cold stress intensity 1–4 [C1–4]) in a fully factorial experimental design, resulting in a total number of 575 plants.

All plants for experiment 1 were transferred from the greenhouse to climate cabinets (Fitotron HGC 1514, Loughborough, UK) with a 12 h light/12 h dark cycle. CO plants were kept in a cabinet at 20 °C, while C1, C2, C3 and C4 plants were kept together in another cabinet at 5 °C for a period of seven days. Within climate cabinets sapling positions were daily randomized, while between cabinets plants and temperatures were interchanged twice during this period. All plants were watered when necessary. At day eight after transfer to the cabinets, C1, C2, C3 and C4 plants were transferred together into a cold chamber (Atmos Chamber MTH-4400, Sanyo Electric Co., Ltd. JP). In the cold chamber, saplings were kept in darkness and exposed to four successive temperature drops at a cooling rate of −5 °C/h, starting at 5 °C. Each temperature drop lasted 1 h and the reached temperature was kept constant for 30 min prior initiation of the next drop. Saplings assigned to different cold stress intensity treatments were successively removed from the cold chamber in between specific cooling steps. C1 plants were removed 30 min after the first temperature drop from 5 °C to 0 °C and the following stress level treatments were removed 30 min after the temperature drops from 0 °C to −5 °C (C2), −5 °C to −10 °C (C3) and −10 °C to −15 °C (C4) (Fig. 2a). After removal from the cold chamber, individuals were immediately transferred to a refrigerator and stored at 4 °C until the next day to allow for controlled thawing. Hence, the thawing temperature and not the thawing rate was standardized for all experimental plants. Subsequently, all saplings (including CO) were transferred back to the greenhouse, where they remained for two weeks at original growing conditions. After this time period, we assessed the proportion of damaged leaves showing signs of chlorosis, necrosis or wilting and the mortality status for each individual by visual inspection (Dataset Cold Stress Injury). The temperature range chosen for this experiment reflects the cold stress intensities experienced by *P. australis* in our study area under field conditions (Supporting information Table S1).

2.4. Experiment 2: metabolic responses to cold stress

In a second batch of *P. australis* saplings, we assessed leaf metabolic responses to a gradient of increasing cold stress intensities. Here, we excluded populations from mid elevational origins for logistic reasons. Five individuals per site × elevational origin combination (only four for site 2-low, Supporting information Table S2) were successively exposed to five experimental treatments (CO, C1, C2, C3, C4) resulting in a total number of 145 plants. The cold stress intensity treatments C1–C4 did not correspond to those used in experiment 1, but consisted of more moderate low temperatures ranging from 5 °C to −4 °C. Based on the

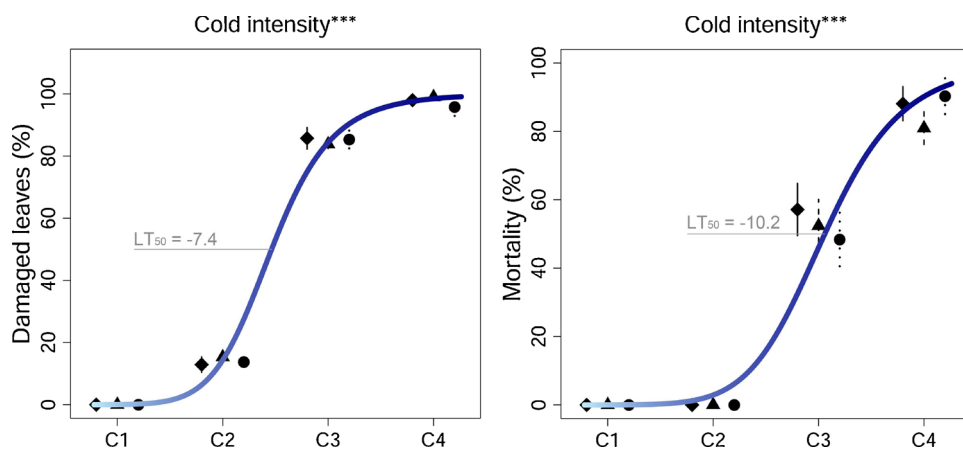


Fig. 3. Cold stress injury. Scatter plots illustrating the effects of cold stress intensity and elevational origin on a) the proportion of damaged leaves and b) mortality in *P. australis*. Symbols represent means and standard errors for low (circles), mid (triangles) and high (rhombuses) elevation populations in cold intensity treatments C1, C2, C3 and C4, while lines represent the model predictions for the effect of cold stress intensity. LT_{50} marks the cold stress intensity at which 50% damage/mortality were reached. Levels of significance based on χ^2 likelihood ratio test for fixed effects terms retained in the minimal adequate LMM are denoted at the top of each plot.

results from experiment 1, we reduced the intensity of cold stress for experiment 2, since we intended to cover a temperature gradient at which metabolic stress responses are effective. Hence, individuals should survive these conditions without damage, which was partially not the case for temperatures lower than -5°C (Fig. 3).

At the age of 105 days (17.02.2018), plants for experiment 2 were transferred from the greenhouse to 20°C (CO) and 5°C (C1–C4) climate cabinets (fabric see experiment 1). During the following seven days, plants were watered and randomized as described for experiment 1. Four hours after the light period had ended at the seventh day, we harvested all leaflets of the four youngest leaves from each plant individual assigned to the CO treatment (20°C cabinet) and C1 treatment (5°C cabinet), flash froze the samples in liquid nitrogen and stored them at -80°C . In the meantime, C2, C3 and C4 plants were transferred to the cold chamber (fabric see experiment 1), where they were further kept in darkness and exposed to three successive temperature drops of $-3^\circ\text{C}/30\text{ min}$, starting at 5°C . Each temperature drop lasted 30 min and the reached temperature was kept constant for additional 30 min prior initiation of the next drop. Plants were harvested 30 min after the temperature drops from 5°C to 2°C (C2), 2°C to -1°C (C3) and -1°C to -4°C (C4) (Fig. 2b). The harvest was conducted in the nighttime hours to reduce circadian effects on metabolism. Saplings showed no signs of injury during leaf harvest and one week after experiment 2 was finished, which supports that the cold stress intensity gradient for experiment 2 included exclusively non-lethal low temperatures.

About 4 mg of lyophilized and homogenized leaf powder per sample were suspended in 715 mL extraction solvent (1 : 2.5 : 1 chloroform [HPLC grade, AppliChem GmbH, Darmstadt, GE] : methanol [LC–MS grade, Geyer GmbH & Co.KG, Hörter, GE] : millipore water) containing ribitol [99%, AppliChem, Darmstadt, GE] as internal standard. Subsequently, 285 mL millipore water were added, samples were vortexed for 15 s, centrifuged for 3 min and supernatants were collected. After drying under a nitrogen flow at 37°C , supernatants were derivatized by sequential methoximation (Alfa Aesar, Karlsruhe, GE) for 90 min and silylation (Macherey-Nagel GmbH & Co. KG, Düren) for 30 min.

Derivatized samples, blanks and an alkane standard mix (C₈–C₄₀; Sigma Aldrich, Karlsruhe, GE) were analyzed on a gas chromatograph (GC) (GC 2010 plus, Shimadzu, Kyōto, JP) equipped with a VF-5 column (30 m $\times$ 0.25 mm i.d.; 10 m guard column; Agilent Technologies, Santa Clara, USA) and coupled with a flame ionization detector (FID) to quantify small (semi-)polar leaf metabolites. Samples were injected at 250°C in a split mode of 1:10 and migrated at a nitrogen-flow of 1.2 mL/min. The temperature program started at 80°C for 3 min followed by a ramp to 310°C at a rate of $5^\circ\text{C}/\text{min}$ and a hold of 3 min. We picked all peaks with a minimum width of 3 s and an area of at least 150 counts from the resulting chromatograms with the GC-Solution Postrun software Version 2.3, (Shimadzu, Kyōto, 2004) and

performed retention time alignment with a maximal linear shift of 0.05 min with the “GC-align” package (Ottensmann et al., 2018) in R version 3.4.4 (R Development Core Team, 2018). We received a total number of 197 metabolic features, which were characterized by their retention indices (RIs) (Kováts, 1958). We calculated the relative concentration of each feature by relating its peak area to the peak area of the internal standard ribitol and to the sample dry weight. For identification of metabolic features, we measured one derivatized sample per population $\times$ temperature treatment combination as well as blanks and an alkane standard mix on a GC equipped with a VF-5 column (same fabrics as for GC-FID analysis) and coupled with a mass spectrometer (MS QP 2020, Shimadzu, Kyōto, JP). The GC-settings corresponded to the analyses described for GC-FID analysis, but using helium as carrier gas. The ion source and transfer line of the MS were set to 230°C and 250°C and analyses were performed in positive electron impact ionization mode at 70 eV. For feature identification, RIs, mass spectra and characteristic qualifier masses were compared to those of commercial references from Sigma-Aldrich (Karlsruhe, GE), Merck (Darmstadt, GE), Roth (Karlsruhe, GE) or Macherey-Nagel (Düren, GE) as well as to the databases of the National Institute of Standards and Technology (Shen et al., 2018) and Golm (Hummel et al., 2010).

Finally, the dataset including the (putative) identities and the relative concentrations of metabolic features was reduced for the statistical analyses. If two features were identified as different analytes of the same metabolite, their relative concentrations determined via GC-FID were summed. Moreover, we removed all metabolites whose mean intensities were lower than twice their mean intensity in the blanks in at least one site $\times$ elevational origin $\times$ cold stress intensity combination. Metabolites that did not occur in at least 50% of the samples in at least one of the site $\times$ elevational origin $\times$ cold stress intensity combinations were also removed. Our final dataset comprised 80 metabolites (Dataset Metabolic Cold Stress Responses).

2.5. Statistical analyses

All statistical analyses were performed with R version 3.4.4 (R Development Core Team, 2018). For data derived from experiment 1, we fitted generalized linear mixed effects models (GLMMs) (R-package “lme4”, Bates et al., 2015), which included either leaf damage (binomial, logit-link) or mortality (binomial, logit-link) as response, elevational origin (factor: low versus mid versus high), cold stress intensity (continuous variable 1–4 corresponding to C1–C4) and their interaction as predictors, as well as site as random effect. Please note that the combination of elevational origin and site corresponds to the population level, which is the reason why population was not included as a random factor. Data for CO plants evidenced that leaf damage and sapling mortality solely resulted from the cold stress intensity treatments (mean $\pm$ SE for both variables = 0 ± 0) and were thus not

further analyzed. After assuring that the maximal models exhibit variance homogeneity and normal distribution of residuals by visual inspection (Zuur et al., 2009), we applied step-wise backward model selection to obtain the minimal adequate models. Here, we removed fixed effect terms with $p \leq 0.05$ based on χ^2 likelihood ratio tests (R-package “MASS”, Venables and Ripley, 2002).

For experiment 2, the relative quantities of the 80 metabolites were compared between CO and C1, C2, C3 or C4 plants within each elevational origin using Mann-Whitney- U tests for nested datasets, which account for the effect of site (R-package “nestedRanksTest”, Scofield, 2015). A two-sided U -test was conducted as described in Schrieber et al. (2019). Moreover, we calculated the fold change in the relative concentration of each metabolite for all elevational origin $\times$ cold stress intensity combinations as *mean relative concentration in C1, C2, C3 or C4 plants / mean relative concentration in CO plants*. We then plotted the p -values resulting from nested Mann-Whitney- U tests on a $\log_{10}$ -scale against the fold changes of each metabolite on a $\log_2$ -scale in volcano plots (Cui and Churchill, 2003). All metabolites had to pass two thresholds to be considered as modulated by cold stress: 1) A p -value ≤ 0.05 (≥ 1.3 on a $\log_{10}$ -scale) and 2) a fold change of either ≤ 0.5 for a decrease of relative concentration (≤ -1 on a $\log_2$ -scale, corresponds to concentration halving) or ≥ 2 for an increase of relative concentration (≥ 1 on a $\log_2$ -scale, corresponds to concentration doubling) under cold stress (Vinaixa et al., 2012).

After determining the metabolites that were modulated by cold stress in *P. australis*, we analyzed whether variation in these induced responses is explained by elevational origin and stress intensity. For this purpose, we focused on modulated metabolites that increased in their relative concentration under cold in at least one of the elevational origin $\times$ cold stress intensity combinations ($\Sigma = 22$ metabolites, Table 2). To account for the possibility that differences in the constitutive amounts of these metabolites may mask or confound differences in induced responses, we fitted linear mixed effects models (LMM), which included the relative concentration of a specific metabolite in CO plants as response, elevational origin (factor: low versus high) as predictors, and site as random effect. In a next step, we calculated fold changes (correspond to induced responses) on the individual level as *relative concentration in C1, C2, C3 or C4 plant individual related to the mean relative concentration in all CO plants from the corresponding elevational origin* after setting all intensities of 0 to the minimum intensity detected across all individuals and all 80 metabolites, namely = 0.027. We then fitted linear mixed effects models (LMMs) that included the $\log_2$ -transformed individual level fold change for each of the 22 metabolites as response, elevational origin (factor: low versus high), cold stress intensity (continuous variable 1–4 corresponding to C1–C4) and their interaction as predictors as well as site as random effect. All of these models were fitted with a maximum likelihood approach. Transformation decisions, model checking, model simplification and significance assessment were applied as described for experiment 1.

3. Results

3.1. Effects of cold stress and elevational origin on leaf damage and mortality

The proportion of damaged leaves and sapling mortality did not depend on the main effect of elevational origin. Instead, our analyses revealed a sharp increase in sapling injury with cold stress intensity (Fig. 3, Table 1) across all three elevational origins. The temperature at which 50% of individuals were injured (LT_{50}) was -7.39°C for leaf damage and -10.19°C for mortality. Interactive effects of elevational origin and cold stress intensity were not observed for either of the injury responses.

3.2. Effects of cold stress on leaf metabolites in low and high elevation populations

Untargeted metabolic fingerprinting and metabolite profiling via GC-FID / MS revealed 27 metabolites that were modulated by cold stress in *P. australis* (Fig. 4, Table 2). Each of these metabolites was modulated in at least one of the eight elevational origin $\times$ cold stress intensity combinations. Of these metabolites, 22 increased in their relative concentration under cold, whereas five decreased in their relative concentration compared to control plants. Twenty-two of the 27 cold stress-modulated metabolites were identified as alcohols, amines and amino acids, sugars and sugar alcohols, organic acids or specialized metabolites with database entries and/or commercial references (Table 2).

We observed substantial differences in induced cold stress responses of *P. australis* between saplings exposed to distinct cold stress intensities and saplings belonging to different elevational origins. Qualitative differences in induced cold stress responses (i.e., identity and number of responding compounds) were addressed with volcano plot analyses (Fig. 4, Table 2), whereas mixed effects models (Fig. 5, Table 1) addressed quantitative differences in induced responses of specific metabolites.

3.2.1. Qualitative differences

Cold stress intensity alone had pronounced effects on the quality of induced stress responses in *P. australis*. When considering samples of both elevational origins together, the number of modulated metabolites increased with cold stress intensity from 15 in C1, over 14 in C2 and 18 in C3 to 21 in C4 (Table 2). The decisive increase in metabolite number took place between C2 and C3, which corresponds to the transition from chilling to frost.

Substantial qualitative differences were also found in induced chemical cold stress responses among *P. australis* populations from low and high elevational origin. When considering all four cold intensities together, 15 metabolites showed a cold-induced increase in the relative concentration in low elevation populations (Fig. 4a, Table 2), whereas 20 metabolites were increased in high elevation populations compared to the unexposed control plants from the respective elevations. Thirteen of these metabolites were induced in both low and high elevation populations (ethanolamine, L-asparagine, L-threonine, γ -aminobutyric acid, cellobiose, fructose, galactinol, glucose, glyceryl glucoside, raffinose, unknown 1, 2, 5). In contrast, adipic acid and xylose only increased in response to cold stress in low elevation populations, whereas catechine, citrate, gallic acid, glutamic acid, L-aspartate, α -ketoglutarate and unknown 4 responded to cold stress exclusively in saplings originating from high elevation populations. Moreover, low elevation populations responded to cold with a reduction in the relative concentration of five metabolites (acetoacetic acid, butanediol, galacturonic acid, palatinose, unknown 3), whereas high elevation populations exhibited no reduction in metabolites in response to cold stress.

Elevational origin and cold stress intensity had not only additive but also interactive effects on the quality of induced cold stress responses in *P. australis*. Here, the number of cold-increased metabolites rose with cold stress intensity in high elevation populations (C1: 9, C2: 9, C3: 13, C4: 18), while this number stayed largely constant across cold intensities for low elevation populations (C1: 11, C2: 9, C3: 12, C4: 11) (Fig. 4, Table 2). The main increase in modulated metabolite number in the saplings from high elevations started between the transition from chilling to frost. High elevation populations additionally induced L-asparagine, citrate, α -ketoglutarate, cellobiose, galactinol, catechine as well as unknown 1, 4 and 5, while low elevation populations additionally increased only L-asparagine and cellobiose. The opposite pattern was observed for metabolites decreasing in concentration in response to cold. The number of these metabolites slightly increased with cold stress intensity in saplings from low elevation populations (C1: 1, C2: 1, C3: 2, C4: 4), while those from high elevation populations

Table 1
Overview of and results for statistical data analyses.

Response	Elevational origin			Cold stress intensity			Elevational origin $\times$ cold stress intensity		
	Est.	<i>p</i>	χ^2	Est. ^{tran.}	<i>p</i>	χ^2	Est.	<i>p</i>	χ^2
Damage	0	> 0.05	0.14	8.56^l	0.00	3102.80	0	> 0.05	1.32
Mortality	0	> 0.05	0.82	8.27^l	0.00	331.60	0	> 0.05	1.02
Ethanolamine	-0.83	< 0.05	4.42	0.43^u	< 0.05	5.84	0	> 0.05	0.48
Glutamic acid	0.98	< 0.001	16.77	0 ^u	> 0.05	0.06	0	> 0.05	1.73
L-Asparagine	0	> 0.05	0.05	0 ^u	> 0.05	3.44	0	> 0.05	2.70
L-Aspartate	0.61	< 0.01	8.51	0 ^u	> 0.05	0.06	0	> 0.05	0.19
L-Threonine	0.94	< 0.01	7.09	-0.36^u	< 0.05	5.32	0	> 0.05	0.29
γ -Aminobutyric acid	0	> 0.05	1.72	0 ^u	> 0.05	0.48	0	> 0.05	0.65
Adipic acid	0	> 0.05	1.12	0 ^u	> 0.05	0.02	0	> 0.05	3.3
Citrate	0	> 0.05	2.09	0.42^u	< 0.001	11.68	0	> 0.05	4.01
Gallic acid	0	> 0.05	3.29	0 ^u	> 0.05	3.35	0	> 0.05	0.12
α -Ketoglutarate	1.51	0.00	15.93	0 ^u	> 0.05	0.59	0	> 0.05	2.29
Cellobiose	0.94	0.04	4.39	0.20^q	< 0.001	23.44	0	> 0.05	0.02
Fructose	0	> 0.05	0.66	0 ^u	> 0.05	0.02	0	> 0.05	0.05
Galactinol	-2.59	i.i.	i.i.	-0.07^q	i.i.	i.i.	0.20	< 0.01	7.16
Glucose	0	> 0.05	0.98	0 ^u	> 0.05	0.40	0	> 0.05	0.08
Glyceryl glucoside	-0.93	0.04	4.10	0 ^u	> 0.05	1.54	0	> 0.05	0.95
Raffinose	0	> 0.05	1.90	0 ^u	> 0.05	1.27	0	> 0.05	1.20
Xylose	0	> 0.05	0.68	0 ^u	> 0.05	0.25	0	> 0.05	0.23
Catechine	0	> 0.05	1.00	0 ^u	> 0.05	2.43	0	> 0.05	2.59
Unknown 1	-2.22	i.i.	i.i.	-0.4^u	i.i.	i.i.	0.77	< 0.05	4.66
Unknown 2	0	> 0.05	2.52	0 ^u	> 0.05	0.71	0	0.38	0.54
Unknown 4	-0.64	i.i.	i.i.	0.02^u	i.i.	i.i.	0.41	< 0.05	4.06
Unknown 5	0	> 0.05	0	0 ^u	> 0.05	0.27	0	> 0.05	0.41

Effects of elevational origin, cold stress intensity and their interaction on leaf damage and sapling mortality (experiment 1) as well as on individual level log₂-fold changes in concentrations of metabolites modulated by cold stress (experiment 2) in *P. australis*. The table provides model parameter estimates (Est.) for all fixed effects terms retained in the minimal adequate (G)LMM for each of the responses as well as *p* and χ^2 -values based on likelihood ratio tests. Bold lettering highlights significant effects. Positive parameter estimates indicate higher response trait values in plants of high compared to those of low elevational origin or an increase in response trait values with increasing cold intensity. The estimate for cold intensity is given with the transformation (^{tran.}) of the term used in the respective model (^l: log₁₀-transformed, ^q: quadratic term, ^u: untransformed).

showed no concentration reduction across cold stress intensity treatments (Fig. 4, Table 2). Again, the main increase in the number of cold-reduced metabolites occurred at the transition from chilling to frost.

3.2.2. Quantitative differences

The analyses of individual level log₂-fold changes with mixed effects models revealed differences in the magnitude of cold-induction between distinct cold stress intensity treatments. The log₂-fold changes of citrate, cellobiose and ethanolamine (Fig. 5e–g) increased with cold stress intensity to a comparable extent in low and high elevation populations (Table 1). While log₂-fold changes for citrate and ethanolamine were positive across all cold intensities, log₂-fold changes of cellobiose became positive only under C4 conditions. The log₂-fold change of L-threonine decreased significantly with increasing cold stress intensity but stayed positive over the entire range of cold intensities (Fig. 5h, Table 1).

Moreover, the analysis disclosed differences in the magnitude of induction for six metabolites between elevational origins. As the relative concentration of cold-modulated metabolites did not differ between control plants from low and high elevation populations except for glutamic acid (Supporting information Table S3), we can largely exclude that differences in constitutive amounts confounded divergence in induced responses. Log₂-fold changes of L-aspartate (Fig. 5a), glutamic acid (Fig. 5b), α -ketoglutarate (Fig. 5d), cellobiose (Fig. 5f) and L-threonine (Fig. 5h) were significantly higher in populations from high than low elevational origin across cold stress intensities (Table 1). Moreover, glyceryl glucoside (Fig. 5c) and ethanolamine (Fig. 5g) showed higher log₂-fold changes in saplings from low compared to those from high elevation populations (Table 1). This general divergence was mainly attributed to data from C1 and C4 plants, but not from C2 and C3 plants (Fig. 5c, g).

The magnitude of induction for three metabolites was significantly affected by the interaction elevational origin $\times$ cold stress intensity.

The log₂-fold changes of galactinol (Fig. 5i) and unknown 1 (Fig. 5j) decreased with increasing cold stress intensity in saplings from low elevations, but increased with cold stress intensity in those from high elevations (Table 1). Unknown 4 exhibited an increase in log₂-fold changes with cold stress intensity in high elevation populations and constant log₂-fold changes across cold stress intensities in low elevation populations (Fig. 5k, Table 1).

4. Discussion

Our experiments provide first support for phenotypic differentiation in plant metabolic cold stress responses in juvenile trees between populations of different elevational origins. In the following, we discuss (1) the potential function of cold-modulated metabolites in the cold stress resistance of evergreen trees; (2) the evolutionary and ecological forces that have potentially driven phenotypic differentiation in metabolic cold stress responses among elevational origins and (3) the implications of our findings for mountain forest restoration.

4.1. Cold resistance in an endemic high mountain tree: from injury estimators to metabolic signature

Leaf damage and mortality of juvenile *P. australis* trees increased with cold stress intensity (experiment 1, Fig. 3, Table 1). The temperature response curves (Fig. 3) suggest that saplings exhibit specific cellular and physiological adaptations to survive moderately low temperatures between 0 and -5°C without incurring severe tissue damage. At the inflection point of curves, these resistance mechanisms start to exhaust, leading to increasing injury as temperatures drop further.

As expected, changes in the concentrations of various primary and specialized metabolites occurred in a temperature range between 5 and -4°C (experiment 2, Fig. 4, Table 2) and likely contributed to the observed cold resistance in *P. australis* saplings. The cold-induced

Table 2
Overview of metabolic cold stress responses.

Compound class	Number	Putative identity	Retention index	Modulated in							
				Low elevation				High elevation			
				C1	C2	C3	C4	C1	C2	C3	C4
Alcohols, amines and amino acids	1	Butanediol	1156				–				
	2	Ethanolamine	1260								
	3	Glutamic acid ^R	1519, 1531	+		+	+	+	+	+	+
	4	L-Asparagine ^R	1593, 1664			+	+				+
	5	L-Aspartate ^R	1422, 1514					+	+	+	
	6	L-Threonine ^R	1292	+	+	+	+	+	+	+	+
	7	γ-Aminobutyric acid ^R	1524	+	+	+	+	+	+	+	
Organic acids	8	Acetoacetic acid	1125		–	–	–				
	9	Adipic acid	1501	+							
	10	Citrate ^R	1827								+
	11	Gallic acid	1948					+		+	+
Sugars and sugar alcohols	12	α-Ketoglutarate ^R	1572							+	+
	13	Cellobiose ^R	2712				+				+
	14	Fructose ^R	1857, 1867	+	+	+	+	+	+	+	+
	15	Galactinol	2961	+	+	+	+				+
	16	Galacturonic acid	1924, 1931	–							
	17	Glucose ^R	1882, 1901	+	+	+	+	+	+	+	+
	18	Glyceryl glucoside	2295	+	+	+	+		+	+	+
	19	Palatinose	2814				–				
	20	Raffinose ^R	3353		+				+	+	
	21	Xylose ^R	1645, 1651			+					
	22	Catechine	2866								+
Specialized metabolites	23	Unknown 1	1580	+	+						+
	24	Unknown 2	2016	+	+	+	+	+	+	+	+
	25	Unknown 3	2481			–	–				
	26	Unknown 4	2857								+
	27	Unknown 5	2900	+	+	+	+	+		+	+

List of metabolites that were significantly (NestedRanks Mann-Whitney-*U* test with $p \leq 0.05$) and pronouncedly (fold change of either ≤ 0.5 for concentration decrease or ≥ 2 for concentration increase) affected by cold stress in at least one elevational origin $\times$ cold intensity treatment combination. Metabolites modulated by cold stress are listed with their compound classes, numbers (corresponding to labels in Fig. 4), putative identities (compounds marked with ^R were identified with commercial references and database entries, the remaining compounds only with database entries), retention indices and direction of modulation (+/– corresponds to concentration increase/decrease under cold).

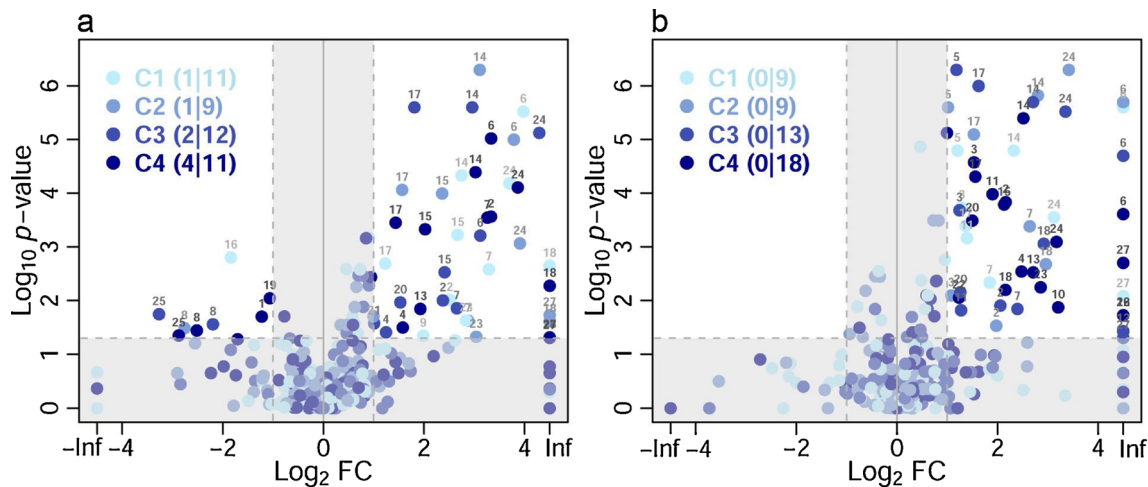


Fig. 4. Metabolic cold stress responses. Volcano plots illustrating the metabolic responses of *P. australis* populations from low (a) and high (b) elevational origins to cold stress treatments with increasing intensity (C1–4, light blue–dark blue). Circles of identical color represent different metabolites corresponding to a specific treatment, with identities numbered consistently with Table 2. The y-axis corresponds to the $\log_{10}$ *p*-values from NestedRanks Mann-Whitney-*U* tests for differences in the relative concentration of metabolites between CO and C1, C2, C3 or C4 plants. The x-axis corresponds to the $\log_2$ fold-change (FC) of metabolites. Compounds with a relative concentration of 0 in the CO/C1–C4 treatment are plotted as $-\text{Inf}/\text{Inf}$ on the $\log_2$ fold change axis. Circles beyond the grey cut off lines represent metabolites modulated by cold stress (i.e., $p\text{-value} \leq 0.05$ [≥ 1.3 on a $\log_{10}$ -scale] and a fold change of either ≤ 0.5 [≤ -1 on a $\log_2$ -scale] for a decrease in relative concentration or ≥ 2 [≥ 1 on a $\log_2$ -scale] for an increase in relative concentration). The number of metabolites modulated by each cold intensity treatment is denoted at the left side of each plot (decrease|increase in relative concentration).

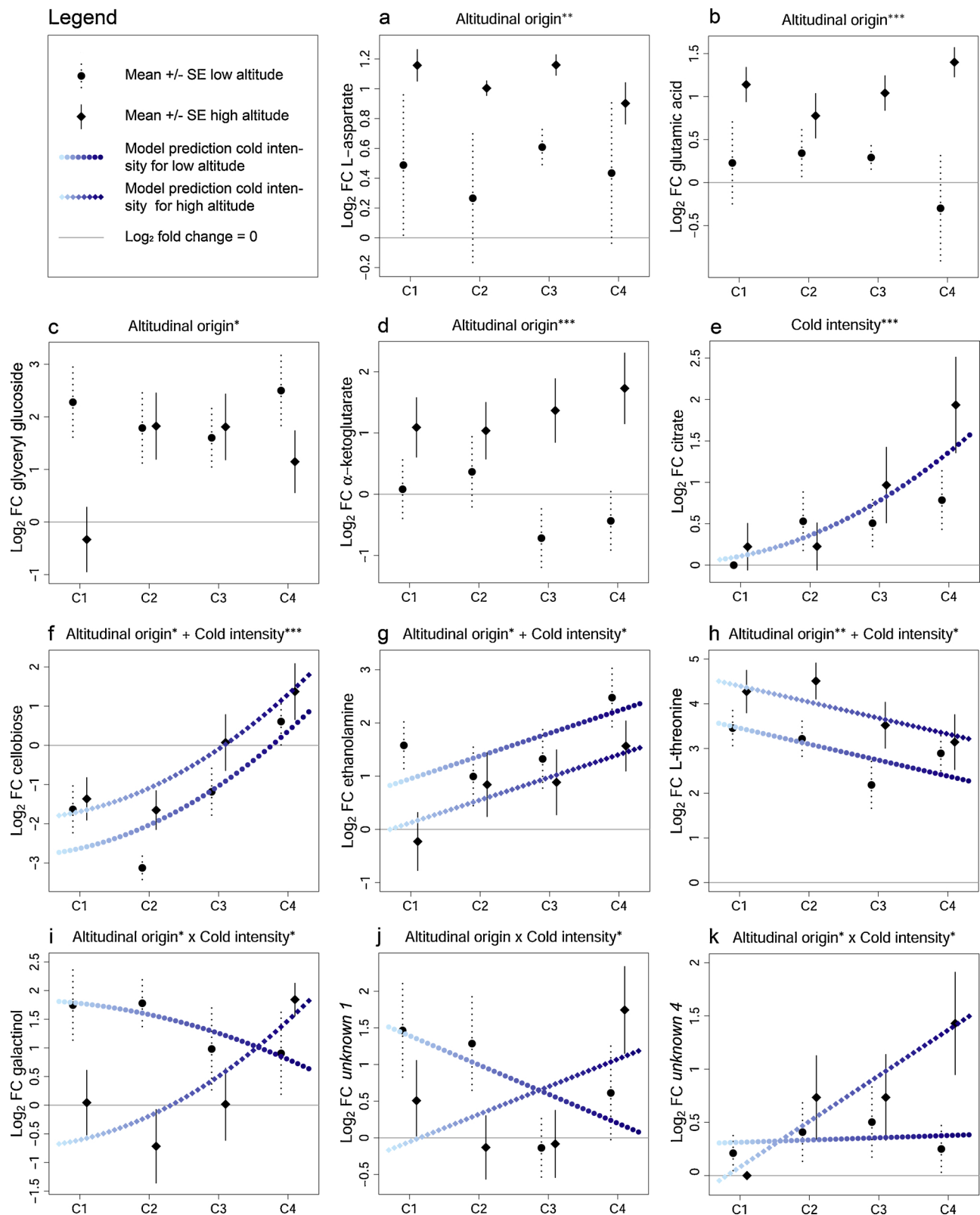


Fig. 5. Magnitude of metabolic cold stress responses. Scatter plots illustrating the effects of elevational origin and cold stress intensity on individual level $\log_2$ -fold changes (FC) in concentrations of metabolites modulated by cold stress in *P. australis*. Vertical solid lines mark a $\log_2$ -fold change of zero. Positive $\log_2$ -fold changes correspond to concentration increases under cold, whereas negative $\log_2$ -fold changes correspond to cold-induced concentration decreases. Symbols represent means and standard errors for low [circles] and high [rhombuses] elevation populations in cold intensity treatments C1, C2, C3 and C4, while dashed lines represent the model predictions for the effect of cold stress intensity in low elevation [dashed with points] and high elevation [dashed with rhombuses] populations. Levels of significance based on χ^2 likelihood ratio test for fixed effects terms retained in the minimal adequate LMM are denoted at the top of each plot (main effect elevation (a–d) and cold intensity (e), as well as additive (f–h) and interactive (i–k) effects of both predictors).

increase of ethanolamine potentially reflects the enhanced synthesis of phosphatidylethanolamine, and thus a compositional change in bio-membranes maintaining their fluidity and stability under cold (Uemura et al., 2006). Decreases in galacturonic acid and acetoacetic acid concentrations indicate alterations in pectine and lipid metabolism potentially related to further cold resistance-mediating membrane modifications (Stumpf and Barber, 1956; Baldwin et al., 2014). The observed increases of sugar, sugar alcohol and amino acid concentrations in *P. australis* have been shown to lower the freezing point of intra- and extra-cellular liquids (Regand and Goff, 2005), to prevent cell-dehydration following extracellular freezing (Kreps, 2002) and to stabilize proteins (Carpenter and Crowe, 1988) as well as bio-membranes (Strauss and Hauser, 1986) under frost drought in other species (reviewed in Sakai and Larcher, 1987; Guy et al., 2007). The detected steep increases in the concentrations of specific proteinogenic amino acids might additionally be related to the induction of lipid-transfer and anti-freeze proteins maintaining the structural integrity of cells under cold stress (Janská et al., 2010), while the induction of the non-proteinogenic γ -aminobutyric acid may be involved in post-translational protein modifications mediating cold resistance (Xin and Browse, 2000). Concentration increases of citrate and α -ketoglutarate indicate enhanced citric cycle activity, which is likely associated with higher energy demands under cold stress. Likewise, a reduction in the concentration of the sucrose analogue palatinose under cold may be related to increased glycolysis for energy production (Börnke et al., 2002). The accumulation of phenolic acids and flavonoids (i.e., gallic acid and catechine) as well as specific sugars and sugar alcohols with high anti-oxidative potential (i.e., raffinose, galactinol) can prevent damage of macromolecules by photosystem-born radicals under cold stress (Yusuf and Toledo, 2004; Kirakosyan et al., 2004; Nishizawa et al., 2008).

Both the number of cold stress-modulated metabolites (Table 2, Fig. 4) and the magnitude of induction (Table 1, Fig. 5e–g) increased with cold stress intensity. Together with the temperature response curves for injury these data suggest that additional metabolic adjustments may occur with time or further temperatures drops.

4.2. Population differentiation in metabolic cold stress responses across an elevational gradient

Leaf metabolic fingerprints revealed clear population divergence related to elevation. Metabolic cold stress responses overlapped for only half of the detected compounds between populations of low and high elevational origins across all cold intensity treatments (Fig. 4, Table 2). Cold-induced increases in concentrations occurred only in high elevation populations for several metabolites with high anti-oxidative potential (gallic acid, catechine), with multiple cryo-protective functions as well as potential involvement in proteinogenic cold responses (glutamic acid, L-aspartate) and compounds related to increased citric cycle activity (citrate, α -ketoglutarate). In contrast, xylose and adipic acid increased exclusively in low elevation populations in a single cold intensity treatment. While xylose has known cryo-protective functions (see above), adipic acid has, to our knowledge, not been reported to mediate plant cold resistance, but to be involved in the degradation of lysine (Arruda et al., 2000). Moreover, the number of compounds with enhanced concentrations under cold increased with cold intensity only in high elevation populations, while quantitative responses of low elevation populations remained relatively constant. Finally, cold-induced concentration decreases, which may not only reflect resistance mediating metabolic changes, but also direct negative effects of cold stress on metabolism (Guy et al., 2007; Heidarvand and Maali, 2010), occurred exclusively in low elevation populations. In summary, these differences support our hypothesis that *P. australis* populations from high elevational origins exhibit more pronounced metabolic cold stress responses with a partially broader reliance on cold stress intensity than populations from low elevations.

In contrast to the metabolite analyses, injury estimators revealed no differences among plants from different elevational origins. However, injury and metabolic responses were assessed at different temperature ranges to evaluate lethal and sub-lethal effects of cold stress (Fig. 2). Because the temperature range for experiment 1 was chosen based on field conditions in our study area (Supporting information Table S1), a lack of elevational differentiation in injury based on unnaturally low or high temperatures is unlikely. Instead, the short time of exposure to cold stress (6 h, excluding the cold acclimation phase) with a relatively fast increase of cold stress intensity can rather explain this finding. It is highly likely that the pronounced differences observed in metabolic stress responses translate into elevational differences in relative growth, leaf damage and mortality under more natural conditions, in which plants are exposed for longer periods of time to repeated freezing/thawing cycles with slower increases in cold stress intensity over weeks. The seemingly contrasting outcomes of experiment 1 and 2 highlight that metabolomics approaches are more sensitive in detecting intra-specific differences in stress susceptibility across small spatial scales than traditionally used injury estimates collected in short-term experiments.

Different mechanisms may underlie the observed phenotypic differentiation in metabolic cold stress responses. It is likely that the higher frequency and intensity of cold stress events towards high elevations selected for increased cold resistance in *P. australis* populations. Next to such local genetic adaptation, random genetic drift and maternal effects mediated by differences in seed provisioning and local heritable epigenetic adjustments may have contributed to the observed population differentiation (Paschke et al., 2003; Gimeno et al., 2008; Chinnusamy and Zhu, 2009). Differences in ploidy levels, which have been found by Kessler et al. (2014) for *P. australis*, have unlikely contributed to the population divergence observed in our study. The majority of our study populations has also been studied by Kessler et al. (2014). For these populations, Kessler et al. (2014) found that *P. australis* individuals are predominantly tetraploid and that variation in ploidy levels is not related to elevation.

Future studies may implement reciprocal transplant experiments using a “sympatric versus allopatric” approach combined with population genetic analyses to disentangle and quantify the contribution of selection, genetic drift, seed provisioning and heritable epigenetic modulations to the differentiation in cold stress responses of *P. australis* (Blanquart et al., 2013; Richards et al., 2017). Such approaches are specifically valuable because they cover the decisive natural diurnal and annual variation in environmental conditions that cannot be mimicked under laboratory conditions. Moreover, they do not only account for a single environmental condition varying across the elevational gradient, but for their combination. The elevational gradients inhabited by the genus *Polylepis* do not only exhibit decreasing temperatures but also increasing UV-B radiation towards the top, whereas drought stress intensifies towards the foothills (Marcora et al., 2017; Cáceres et al., 2019). While the physiological response to each of these stressors has been revealed to a considerable extent when applied individually, their interactive effects on the metabolome are less well understood (Mittler, 2006; Flexas and Gago, 2018). Future studies using *Polylepis* spp. as models may generate valuable information on the evolution of cross-resistance to multiple stressors with a focus on overlapping and non-overlapping components of the metabolic response as well as on trade-offs and synergistic interactions among specific responses. Studies of this kind could further characterize tree functional responses to relevant environmental conditions from an integrative methodological perspective to enhance our understanding of forest ecology for the improvement of future conservation projects. Moreover, molecular studies employing a ‘new generation of model species’ from extreme environments for the study of plant $\times$ environment interactions will broaden our knowledge on plant functioning beyond *Arabidopsis thaliana* and crop plants (Flexas and Gago, 2018).

4.3. Implications for mountain forest restoration

As *Polylepis* mountain forests experienced severe degradation during the past decades (IUCN, 2017), restoration measures are necessary to prevent further population extinctions and regenerate impaired stands. Such measures typically use seeds, seedlings or saplings (Rorato et al., 2018) - life stages that are known to be most sensitive to environmental stress (Hofmann et al., 2014). Numerous studies on herbaceous plants demonstrated that restoration efforts obtain better results with local provenancing, i.e., the use of regional seed sources (Weißhuhn et al., 2012; Bucharova et al., 2017), because plants are often adapted to their local environment (Leimu and Fischer, 2008). Our results indicate that stress resistance in *P. australis* is related to the environmental conditions prevailing at the elevational origins of populations, implying that the selection of seed material for forest restoration should orient towards these relatively small spatial scales to be at the safe site. It has to be noted that our findings can only be generalized for the predominantly tetraploid individuals in the northern parts of the Córdoba mountains, as diploid north-western Andean and southern Córdoba mountain populations with mixed ploidy levels (Kessler et al., 2014) may exhibit different relationships among stress resistance and elevational origin. However, the pronounced variation in ploidy levels further highlights the need for local provenancing in the sustainable restoration of *P. australis* forests.

Author contributions

KS, YC and CM planned and designed the study. YC, PM and DR conducted the field sampling. YC raised the experimental plants and assessed plant performance in experiment 1. KS and YC conducted experiment 2 and KS and AE performed the chemical analyses. KS performed the statistical analyses. KS and YC wrote the first version of the manuscript. KS revised the manuscript and CM, YC, IH and DR contributed to the final version. IH, PM and DR acquired funding.

Declaration of Competing Interest

The authors do not have any financial or personal relationships with other people or organizations that could have inappropriately influenced or biased the study.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.envexpbot.2019.103918>.

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